

DL for ENG

LECTURE 12.1

Power Grid Stability Estimation

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GRID · STATE · INERTIA



A grid is a graph with the power flow equations at every bus, and a control room sees only a few noisy meters of it. The lecture estimates the full state with the estimator every utility runs and with a network judged by the physics, with no true state to train on - then lets the grid move: the swing equation, and a plant's inertia recovered from a recorded disturbance.

Learning Objectives

SECTIONS, IN ORDER

- 1 **The grid** — a graph, the power flow equations, what a room sees
- 2 **State estimation** — least squares, and a network with no true state
- 3 **The swing** — the equation of motion, and the inertia from a record

UDL carries the graph network, Liu the inverse fit. The example is Ex_12.1's six-bus grid, representative of eastern Denmark and not a model of it: the line impedances are textbook values, because no operator publishes measured ones, and results that depend on them must say so.

READ, AND WHAT EACH ANSWERS

UDL ch. 13	how is a graph represented?
Liu ch. 4	what can readings determine?
SciPy	how is an ODE shot and fitted?

Recap of L11.2 Dynamics and Energy

1 – 2 • VEHICLE MODEL

1 What is the kinematic bicycle model, and what must be measured before it is used?

The car reduced to two wheels: speed and steering angle in, position and heading out, with the wheelbase L as its only parameter. The map from steering command to angle is not linear and is calibrated by measurement, or every later result is in doubt. *Ex_11.2 on the car*

2 What is the grip limit, and why is it enforced hard?

The tyres give one budget of force, which braking and cornering share, so the speed in a bend of radius R cannot pass $\sqrt{(\mu g R)}$. A penalty is a preference the optimiser may trade away; a bounded layer cannot be traded, and on a car the difference is a crash. *Ex_11.2 section 5; L7.1*

3 – 4 • ENERGY

3 Where does the energy go, and why is the cheapest lap not the slowest?

Into winding loss, which grows with the square of the current, into rolling resistance, and into the computer, which draws a constant power for as long as the lap takes. A slow lap pays the constant load longer, so the optimum lies in between, at a smooth speed profile. *Ex_11.2 sections 4 – 5*

4 How are the model's parameters identified, and what makes the result checkable?

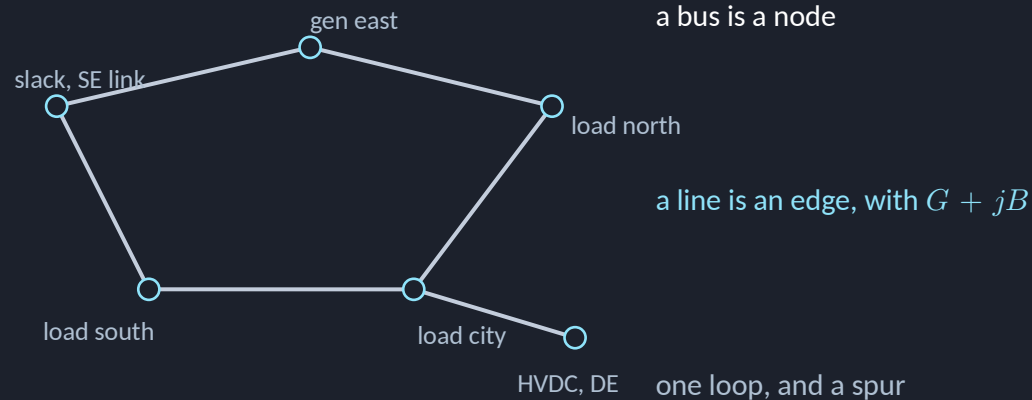
From an excitation run with steps, ramps and a coast-down; a steady cruise cannot separate the armature resistance from the rolling resistance, though the fit looks good. The current sensor measures the energy of a lap, so the prediction is compared with a measurement. *Ex_11.2 section 6; L10.1*

A Grid Is a Graph

Transmission grid — buses joined by lines: a graph, with a node's state being its voltage magnitude and angle, and every edge carrying an admittance. The state of the whole grid is one number pair per bus, and everything a control room wants to know is a function of it.

EX_12.1'S SIX BUSES, AS THEY ARE WIRED

SIX BUSES, SIX LINES, ONE LOOP



THE CASE, AND ITS LIMITS

the grid 6 buses, 6 lines, 100 MVA base

the state 6 magnitudes, 5 angles

stands for eastern Denmark, loosely

impedances textbook values, flagged

so say when a result leans on them

No operator publishes measured impedances; representative is the honest word.

The Power Flow Equations

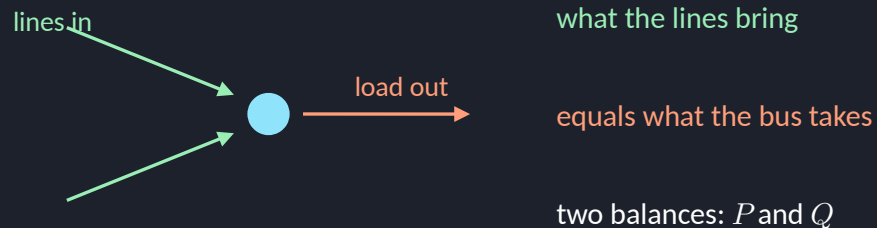
Power flow equations — two algebraic balances at every bus: the active and the reactive power the lines bring must equal what the bus injects or takes. They are constraints the voltages satisfy at every instant, not a simulation in time - the elliptic habit of L8.1, on a graph.

TWO BALANCES AT EVERY BUS

$$P_i = V_i \sum_k V_k (G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik}), \quad Q_i = V_i \sum_k V_k (G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik})$$

V_i, θ_i the voltage magnitude and angle at bus i : the state, G, B the admittance matrix, $\theta_{ik} = \theta_i - \theta_k$

ONE BUS, ONE BALANCE



HOW THE EXERCISE USES THEM

as a solver	Newton's method: the reference
as a check	the residual at every bus
as a loss	the term that replaces truth
in torch	so autograd reaches the state

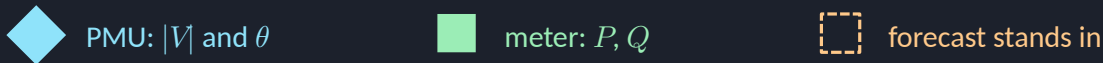
The same equations serve three ways, and the third is this lecture's turn: a residual can judge an estimate at a bus no meter sees.

What the Control Room Sees

Measurement model — every reading is a function of the state plus noise: a PMU reads a bus's magnitude and angle, a meter its P and Q , a forecast stands in where no meter exists. The thin set has fewer readings than unknowns, and where they sit on the graph decides everything.

ONE LINE PER METER

$z_j = h_j(V, \theta) + \varepsilon_j$
 z_j one reading, h_j what that meter measures as a function of the state, ε_j its noise



TALK • 2 MIN How would you explain, in your own words, why a grid's state can be known better than its meters? Then, from the table on the right, why does the thin set fail where the forecast set does not?

THE THREE SETS OF EX_12.1

set	readings	covers
full	14	every unknown
thin	6	5 of 11 unknowns
forecast	6 + priors	every unknown, loosely

The forecast's accuracy, 0.05 p.u., is an assumption of the exercise. Observability is a property of where the meters sit on the graph, not of how good they are: the thin set leaves half the state unseen however exact its six readings become.

Weighted Least Squares

State estimation — finding the voltage magnitude and angle at every bus from fewer, noisy readings. The estimator every utility runs is weighted least squares: each meter's misfit, weighted by its accuracy, minimised by Newton's method. It is the baseline any network must face.

THE ESTIMATOR IN SERVICE

$$\hat{x} = \arg \min_x \sum_j \frac{(z_j - h_j(x))^2}{\sigma_j^2}$$

$\hat{x}$ the estimated state, σ_j each meter's accuracy : a good meter weighs more, solved by Newton's method

THE THIN SET: 5 OF 11 UNKNOWNNS



determined by the six readings

unseen, whatever the noise

MEASURED, OVER 300 OPERATING POINTS

set	worst V	angle
full, 14	0.0030 p.u.	0.07°
thin, 6	0.0248 p.u.	1.82°
forecast	0.0050 p.u.	0.15°

One estimate costs about a millisecond. The thin row is not a solver failure: it is the graph saying no.

Judged by the Physics

Estimation with no true state — a network maps the readings to the state, and its loss has no true state in it: the misfit to the meters, plus the power flow residual at every bus. The physics constrains the buses no meter sees, and the same residual is the estimate's own health check.

THE LOSS, WITH NOTHING TO COPY

$$\mathcal{L} = \sum_j \frac{(z_j - h_j(\hat{x}))^2}{\sigma_j^2} + w_P \sum_i r_i(\hat{x})^2, \quad \text{no true state in it}$$

r_i the power flow residual at bus i : physics constrains every bus, metered or not, w_P its weight

THE LOSS IN PYTORCH

```
net = MLP(n_in=14, n_hidden=64, n_layers=3)    # readings in, the state out
V, th = net(z)                                # 9995 weights in all
r = power_flow_residual(V, th)                 # the physics, in torch
L = wls_misfit(z, V, th) + W_P * r.pow(2).sum() # no true state anywhere in it
```

AGAINST THE BASELINE, HONESTLY

accuracy 0.0043 p.u.: it does not win

one estimate 0.26 ms; 2 μ s in a batch

training 13 s, once

a line out, untold both fail alike

but the check its residual: 0.28 to 2.35

The gain is the time of one estimate, and a self-check that says when not to trust it.

The Swing Equation

Swing equation — the grid set in motion: a machine's rotor accelerates with the gap between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together. One ordinary differential equation per machine.

THE EQUATION OF MOTION

$$\frac{d\delta}{dt} = \omega - \omega_s, \quad \frac{2H}{\omega_s} \frac{d\omega}{dt} = P_m - P_e(\delta) - D \frac{\omega - \omega_s}{\omega_s}$$

δ the rotor angle, ω its speed, H the inertia in seconds, D damping, P_m, P_e mechanical and electrical power

AFTER A FAULT



the rotor swings, and settles

H sets how slowly

D sets how it dies away

WHY INERTIA IS THE NUMBER

H seconds of stored spin

falling converters replace machines

so the grid swings faster

and operators must know today's H

from what the disturbances it records

DK2 swings with the Nordic area, so the Nordic frequency is the right record for a DK2 plant.

The Inertia From a Record

Inertia identification — the inverse problem of the block: H and D trained beside the weights so the swing equation fits a recorded disturbance, as L8.2's diffusivity was. The residual is written with H on the left, so a large H cannot hide, and only the record after the fault is used.

THE FIT, WRITTEN SO IT CANNOT HIDE

$$H, D \text{ trainable, } \mathcal{L} = \left\| \frac{2H}{\omega_s} \dot{\omega} - P_m + P_e + D \Delta\omega \right\|^2 + w_D \mathcal{L}_{\text{record}}$$

$\mathcal{L}_{\text{record}}$ the misfit to the recorded frequency, after the fault is cleared; H on the left, so a large H cannot hide

THE FIT IN PYTORCH

```
H = torch.nn.Parameter(torch.tensor(8.0))      # trained with the weights
res = (2*H/w_s)*grad(w, t) - P_m + P_e + D*dw   # H on the left
L = res.pow(2).mean() + W_D*(w - rec).pow(2).mean() # after the fault
```

TALK • 2 MIN How would you decide, in your own words, which part of a record identifies a parameter? Then, from the table on the right, why does the window before the fault fail from both starts?

THE RECORD WAS MADE WITH $H = 4.00$ s

fit	start 8 s	start 2 s
shooting, after	7.41 s	1.83 s
physics-informed, after	3.89 s	3.90 s
shooting, before	50 s, the bound	1.94 s
physics-informed, before	8.35 s	2.10 s

Shooting has local minima on this record; the physics-informed fit did not - a finding about this record and these starts, not a law. Before the fault nothing excites the machine, and no method identifies what the data does not carry.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
bad-data detection	a wrong meter, found by its residual	Liu ch. 4	every estimator in service
<u>placing one more meter</u>	where a reading buys observability	pandapower docs	the thin set's missing half
<u>the critical clearing time</u>	how long a fault may stay	PowerSystems.jl	protection settings
<u>rate of change of frequency</u>	the first seconds of a real record	ENTSO-E data	low-inertia grids
the graph network's layers	messages passed along the lines	UDL ch. 13	grids too large for one MLP; L12.2

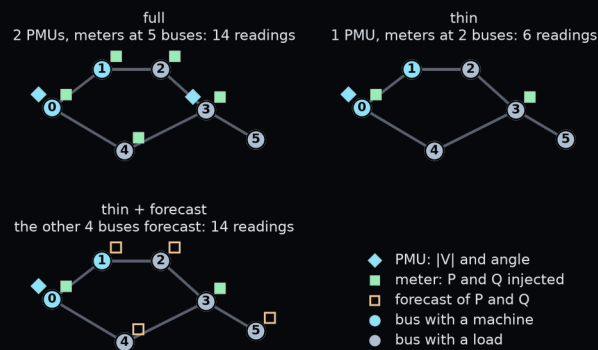
WHY THEY ARE LEFT OUT

Ex_12.1 estimates one six-bus grid from fourteen readings and fits one machine's inertia, scored against Newton's power flow, so none of them is needed to do it. The last row is where the next lecture begins.

Exercise 12.1

ONE NOTEBOOK, SECTION BY SECTION

- 1 – 3 — the grid, the three sets of readings, the power flow
- 4 — Newton's reference; weighted least squares on each set
- 5 — the network judged by the physics; then a line out, untold
- 6 — the inertia from the recorded disturbance, two ways
- 7 – 9 — the report; mini projects MP12.1A and MP12.1B



THE QUESTION EACH PART ANSWERS

- 1 – 3 what is the state of a grid?
- 4 how far do the meters carry?
- 5 what replaces the true state?
- 6 what does a record identify?

REPORT QUESTION, TO CONCLUDE

The network matched least squares from the same readings and answered three hundred estimates in a blink; its residual rose eightfold when a line went out untold. In a control room, what would you use each tool for?

Key Takeaways

STATE ESTIMATION

what it is	the state from few, noisy meters
function	every bus's V and θ
train, tune	no true state: physics in the loss
vs least squares	equal accuracy, a fraction of the time
used for	a room's picture, and its self-check

SWING DYNAMICS

what it is	the grid's equation of motion
function	angles and speeds after a fault
train, tune	H, D beside the weights
vs shooting	no local minima on this record
used for	today's inertia, from records

Next — L12.2: whether the grid survives the next fault, predicted before it happens.

Questions

1 – 2 • STATE ESTIMATION

1 What are the power flow equations, and what is state estimation?

Two algebraic balances at every bus, of active and of reactive power, written with the admittance matrix. They are constraints the voltages must satisfy, not a simulation in time. State estimation finds the voltage magnitude and angle at every bus from fewer, noisy measurements.

Ex_12.1 sections 3 – 4

2 What does observability depend on, and what does a physics-informed graph network add?

On where the measurements sit in the graph, not on how good the sensors are. The network outputs the state and is judged by the power flow residual at every bus, so a bus with no meter is still constrained. Weighted least squares stays the baseline to beat. *Ex_12.1 sections 4 – 5; L5.1*

3 – 4 • SWING DYNAMICS

3 What does the swing equation add to power flow?

Time. The rotor angle of each machine accelerates with the difference between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together through the grid. It is an ordinary differential equation in the loss. *Ex_12.1 section 6*

4 How are inertia and damping identified, and how is an estimate checked with no ground truth?

They are trained beside the weights so that the swing equation fits a recorded frequency disturbance, as the diffusivity was in L8.2. With no true state to compare against, the check is the residual map: where the physics is not satisfied, the estimate is not to be trusted. *Ex_12.1 sections 5 – 6; L8.2*

NEXT

L12.2 — Power Grid Stability Prediction

From what the grid is to what it will do: whether it survives the next fault, predicted by a graph network in the time a protection relay has.

BEFORE THE NEXT SESSION

read	UDL ch. 13
run	Ex_12.1, sections 1 to 6
bring	your inertia, from both starts
think about	what the residual map told you

The estimation half of the block is done: the state, and the machine behind it.